

S26 — Probability: Multiplication Rule & Conditional Probability

Data/values vertical format mein hain. No hints.

1. Multiplication Rule — Independent Events

A fair coin is tossed twice.

Answer:

1. Write the complete sample space.
 2. Define Event A = first toss is Head.
 3. Define Event B = second toss is Head.
 4. Find .
 5. Find .
 6. Find .
 7. Are A and B independent?
 8. Explain why.
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2. Multiplication Rule — Dice

A fair die is rolled twice.

Answer:

1. Probability of getting a 6 on the first roll.
2. Probability of getting a 6 on the second roll.
3. Probability of getting 6 on **both** rolls.
4. Are the two rolls independent?
5. What happens to the probability if the die is rolled three times and we want 6 every time?

3. Conditional Probability

A company has 100 employees.

Department	Python Users	Non-Python Users
Data Analyst	30	10
Finance	15	45

Answer:

1. Total number of employees.
2. Total Python users.
3. Probability that a randomly selected employee is a Python user.
4. Probability that an employee is a Data Analyst.
5. Probability that an employee is a Python user **given that they are a Data Analyst.**
6. Probability that an employee is a Data Analyst **given that they are a Python user.**
7. Are “Data Analyst” and “Python User” independent? Explain.

4. Conditional Probability — Customer Behavior

A company has 200 customers.

Customer	Purchased	Did Not Purchase
Saw Ad	80	40
Did Not See Ad	20	60

Answer:

1. Total customers who saw the advertisement.
2. Total customers who purchased.

3. Find .
 4. Find .
 5. Find .
 6. Which probability is larger: Q4 or Q5?
 7. Explain why these two conditional probabilities are not the same.
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5. Independent or Dependent?

Identify whether the events are **independent or dependent**.

1. Tossing a coin and rolling a die.
2. Selecting a card, replacing it, then selecting another card.
3. Selecting a card without replacement, then selecting another card.
4. Rain occurring and whether an umbrella is carried.
5. A customer's age and their date of birth.

Give a short reason for each.

6. Find the Wrong Interpretation

An analyst says:

“40% of customers purchased after seeing an advertisement. Therefore, there is a 40% probability that every customer who sees the advertisement will purchase.”

Answer:

1. Is this conclusion correct?
 2. What is wrong with the statement?
 3. What does the 40% actually represent?
 4. If 500 customers see the advertisement, approximately how many might purchase?
 5. Is that number guaranteed?
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7. Multiplication Rule — Business Case

A website has:

- $P(\text{Visitor clicks Ad})=0.30$
- $P(\text{Visitor purchases}|\text{Clicked Ad})=0.20$

Answer:

1. What does the second probability mean?
 2. Calculate the probability that a visitor **clicks the ad and purchases**.
 3. Out of 10,000 visitors, approximately how many would both click and purchase?
 4. Is this an independent-events situation?
 5. Why?
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8. Conditional Probability vs Joint Probability

A dataset contains 1,000 customers.

- 300 customers used the mobile app.
- 120 customers used the mobile app and purchased.
- 400 customers purchased.

Answer:

1. Find .
 2. Find .
 3. Find .
 4. Find .
 5. Explain the difference between Q3 and Q4.
 6. Which probability is useful for answering: **“Among app users, how many purchase?”**
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9. Find the Wrong Analyst Conclusion

An analyst observes:

- $P(A)=0.50$
- $P(B)=0.40$
- $P(A \text{ and } B)=0.20$

They say:

“Because , A and B must be independent.”

Answer:

1. Is the conclusion correct?
 2. What condition is being checked?
 3. Based on the given values, are A and B independent?
 4. Explain your answer.
 5. What would need to equal for independence?
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10. Full Data Analyst Case

An e-commerce company analyzes **1,000 customers**.

Customer Group	Purchased	Did Not Purchase
Saw Advertisement	240	160
Did Not See Advertisement	60	540

Answer:

1. Total customers who saw the advertisement.
2. Total customers who purchased.
3. $P(\text{Purchased})$
4. $P(\text{Purchased} \mid \text{Saw Advertisement})$
5. $P(\text{Purchased} \mid \text{Did Noy See Advertisement})$

6. $P(\text{Saw Advertisement and Purchased})$
7. Is seeing the advertisement associated with purchasing?
8. Calculate the difference between the two conditional purchase probabilities.
9. Can you conclude that the advertisement **caused** the purchases? Explain carefully.
10. Give one business recommendation based on the results.